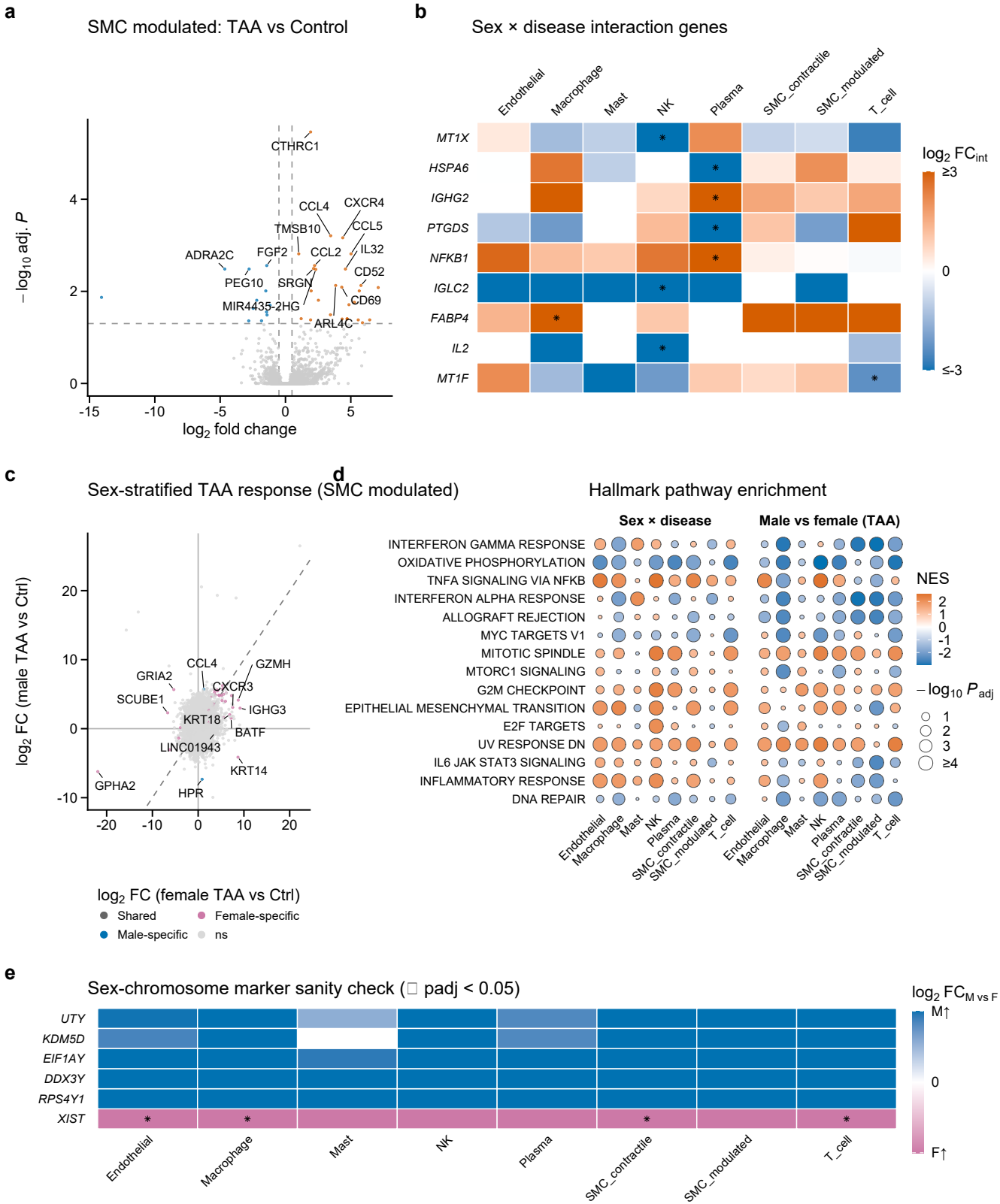

a
SMC modulated: TAA vs Control
CTHRC1
CCL4
CXCR4
TMSB10
CCL5
ADRA2C
FGF2
CCL2
IL32
CD52
PEG10
SRGN
CD69
MIR4435-2HG
ARL4C
−log10 adj. P
log2 fold change
b
Sex × disease interaction genes
Endothelial
Macrophage
Mast
NK
Plasma
SMC_contractile
SMC_modulated
T_cell
MT1X
HSPA6
IGHG2
PTGDS
NFKB1
IGLC2
FABP4
IL2
MT1F
log2 FCint
≥3
0
≤-3
c
Sex-stratified TAA response (SMC modulated)
log2 FC (male TAA vs Ctrl)
GRIA2
CCL4
GZMH
CXCR3
SCUBE1
IGHG3
KRT18
BATF
LINC01943
KRT14
GPHA2
HPR
log2 FC (female TAA vs Ctrl)
Shared
Female-specific
Male-specific
ns
d
Hallmark pathway enrichment
Sex × disease
Male vs female (TAA)
INTERFERON GAMMA RESPONSE
OXIDATIVE PHOSPHORYLATION
TNFA SIGNALING VIA NFKB
INTERFERON ALPHA RESPONSE
ALLOGRAFT REJECTION
MYC TARGETS V1
MITOTIC SPINDLE
MTORC1 SIGNALING
G2M CHECKPOINT
EPITHELIAL MESENCHYMAL TRANSITION
E2F TARGETS
UV RESPONSE DN
IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
DNA REPAIR
NES
−log10 Padj
e
Sex-chromosome marker sanity check (□ padj < 0.05)
UTY
KDM5D
EIF1AY
DDX3Y
RPS4Y1
XIST
log2 FCM vs F
M↑
F↑